CIS 5512 – Homework 3

Md. Mizanur Rahman
md.mizanur.rahman@temple.edu

Task 1

Let 10 seconds be the measured sequential running time and 40 seconds be the parallel running time using 10 processors for a parallel application. Predict the speedup for 30 processors using Amdahl's Law and Gustafson's Law.

Given

$$P = 10$$

$$T_{\text{seq}}(10) = 10 \text{ sec}$$

$$T_{\text{par}}(10) = 40 \text{ sec}$$

The total execution time on 10 processors is

$$T(10) = 10 + 40 = 50 \text{ sec.}$$

Baseline: Gustafson's Law at $P = 10$

The serial fraction measured on the 10-processor system is

$$\beta_g(10) = \frac{T_{\text{seq}}(10)}{T_{\text{seq}}(10) + T_{\text{par}}(10)}$$
$$= \frac{10}{10 + 40} = \frac{10}{50} = 0.2.$$

Therefore, the parallel fraction is

$$1 - \beta_g = 1 - 0.2 = 0.8.$$

Using Gustafson's Law,

$$S_g(P) = \beta_g + (1 - \beta_g)P.$$

For 10 processors,

$$S_g(10) = 0.2 + (1 - 0.2)(10)$$

$$= 0.2 + 0.8(10)$$

$$= 0.2 + 8 = \boxed{8.2}.$$

Baseline: Amdahl's Law at $P = 10$

The parallel part takes 40 seconds using 10 processors.

Its equivalent execution time using one processor is

$$T_{\text{par}}(1) = 40 \times 10 = 400 \text{ sec.}$$

The sequential portion remains

$$T_{\text{seq}}(1) = 10 \text{ sec.}$$

Therefore, the equivalent total execution time on one processor is

$$T(1) = 10 + 400 = 410 \text{ sec.}$$

The Amdahl serial fraction is therefore

$$\beta_a = \frac{10}{410} = 0.02439.$$

Using Amdahl's Law,

$$S_a(P) = \frac{1}{\beta_a + \frac{1-\beta_a}{P}}.$$

For 10 processors,

$$S_a(10) = \frac{1}{0.02439 + \frac{1-0.02439}{10}}.$$

$$= \frac{1}{0.02439 + 0.097561}.$$

$$S_a(10) \approx \boxed{8.2}.$$

Thus, both laws produce the same baseline speedup:

$$\boxed{S_g(10) = S_a(10) = 8.2}.$$

Increase the Number of Processors: $10 \rightarrow 30$

Gustafson's Law: Open/Scaled Problem Size

For an open or scaled problem, the measured serial fraction is maintained:

$$\beta_g = 0.2.$$

For 30 processors,

$$S_g(30) = 0.2 + (1 - 0.2)(30).$$

$$= 0.2 + 0.8(30).$$

$$= 0.2 + 24.$$

Therefore,

$$\boxed{S_g(30) = 24.2}.$$

Gustafson's Law: Fixed Problem Size

If the problem size remains fixed, increasing the number of processors from 10 to 30 gives three times as many processors.

Therefore, the parallel execution time becomes

$$T_{\text{par}}(30) = \frac{40}{3} = 13.333 \text{ sec.}$$

The sequential execution time remains

$$T_{\text{seq}}(30) = 10 \text{ sec.}$$

The new Gustafson serial fraction is

$$\beta_g(30) = \frac{10}{10 + 13.333}.$$

$$= \frac{10}{23.333} \approx 0.42857.$$

Using Gustafson's Law,

$$S_g(30) = 0.42857 + (1 - 0.42857)(30).$$

$$= 0.42857 + 0.57143(30).$$

$$= 0.42857 + 17.1429.$$

Therefore,

$$\boxed{S_g(30) \approx 17.57}.$$

Amdahl's Law: Fixed Problem Size

For a fixed-size problem, the Amdahl serial fraction remains

$$\beta_a = \frac{10}{410} = 0.02439.$$

For 30 processors,

$$S_a(30) = \frac{1}{0.02439 + \frac{1-0.02439}{30}}.$$

$$= \frac{1}{0.02439 + \frac{0.97561}{30}}.$$

$$= \frac{1}{0.02439 + 0.03252}.$$

$$= \frac{1}{0.05691}.$$

Therefore,

$$\boxed{S_a(30) \approx 17.57}.$$

Amdahl's Law: Open/Scaled Problem Size

For the open-size case, the parallel execution time remains approximately 40 seconds while the number of processors increases to 30.

The equivalent parallel work on one processor is

$$T_{\text{par}}(1) = 40 \times 30 = 1200 \text{ sec.}$$

The sequential part remains

$$T_{\text{seq}} = 10 \text{ sec.}$$

Therefore, the new Amdahl serial fraction is

$$\beta_a = \frac{10}{10 + 1200}.$$

$$= \frac{10}{1210} = 0.008264.$$

Using Amdahl's Law,

$$S_a(30) = \frac{1}{0.008264 + \frac{1-0.008264}{30}}.$$

$$= \frac{1}{0.008264 + \frac{0.991736}{30}}.$$

$$= \frac{1}{0.008264 + 0.033058}.$$

$$= \frac{1}{0.041322}.$$

Therefore,

$$\boxed{S_a(30) = 24.2}.$$

Final Results for Question 1

Case	**Gustafson**	**Amdahl**
Baseline, $P = 10$	8.2	8.2
Fixed size, $P = 30$	17.57	17.57
Open/scaled size, $P = 30$	24.2	24.2

Therefore, for the fixed-size problem,

$$\boxed{S(30) \approx 17.57}.$$

For the open/scaled-size problem,

$$\boxed{S(30) = 24.2}.$$

Task 2

Explanation of the Projection Difference

Amdahl's Law predicts a speedup of approximately 17.57 using 30 processors, whereas Gustafson's Law predicts 24.2. The difference arises from the workload assumptions. Amdahl's Law is conventionally applied to a fixed workload, where the sequential component limits the performance improvement as processors increase. Gustafson's Law considers a scaled workload, where additional processors allow a larger amount of parallel work to be completed while maintaining approximately the same parallel execution time and measured serial fraction. Therefore, Gustafson's Law predicts a higher speedup because it accounts for increasing the amount of useful parallel work. The two laws are mathematically consistent when applied to the same workload. If Gustafson's serial fraction is recalculated for the unchanged fixed workload, it also produces 17.57.

Task 3

Speedup with 100 and 200 Processors

Using 100 processors, the sequential part is 10% of the total 150-second running time of a parallel program. Find the speedup using Gustafson's Law. Then predict the speedup for 200 processors using Amdahl's formula.

Given

$$P = 100$$

$$T(100) = 150 \text{ sec}$$

The sequential portion is

$$10\%.$$

Therefore,

$$T_{\text{seq}}(100) = 0.10(150) = 15 \text{ sec}.$$

The parallel execution time is

$$T_{\text{par}}(100) = 150 - 15 = 135 \text{ sec}.$$

Gustafson's Law for 100 Processors

The Gustafson serial fraction is

$$\beta_g(100) = \frac{15}{150} = 0.1.$$

Using Gustafson's Law,

$$S_g(100) = 0.1 + (1 - 0.1)(100).$$

$$= 0.1 + 0.9(100).$$

$$= 0.1 + 90.$$

Therefore,

$$\boxed{S_g(100) = 90.1}.$$

Convert to Amdahl's Serial Fraction

The parallel portion takes 135 seconds using 100 processors.

Its equivalent one-processor execution time is

$$T_{\text{par}}(1) = 135 \times 100 = 13500 \text{ sec.}$$

The sequential execution time remains

$$T_{\text{seq}}(1) = 15 \text{ sec.}$$

Therefore, the total equivalent one-processor execution time is

$$T(1) = 15 + 13500 = 13515 \text{ sec.}$$

The Amdahl serial fraction is

$$\beta_a = \frac{15}{13515}.$$

$$\beta_a \approx 0.0011099.$$

Amdahl's Law for 200 Processors

For a fixed-size workload,

$$S_a(200) = \frac{1}{\beta_a + \frac{1-\beta_a}{200}}.$$

Substituting,

$$S_a(200) = \frac{1}{0.0011099 + \frac{1-0.0011099}{200}}.$$

$$= \frac{1}{0.0011099 + \frac{0.9988901}{200}}.$$

$$= \frac{1}{0.0011099 + 0.00499445}.$$

$$= \frac{1}{0.00610435}.$$

Therefore,

$$\boxed{S_a(200) \approx 163.82}.$$

Verification Using Gustafson's Law for Fixed Size

The number of processors increases from 100 to 200.

For the same amount of parallel work, the parallel execution time becomes

$$T_{\text{par}}(200) = \frac{135}{2} = 67.5 \text{ sec.}$$

The sequential time remains

$$T_{\text{seq}}(200) = 15 \text{ sec.}$$

The new Gustafson serial fraction is

$$\beta_g(200) = \frac{15}{15 + 67.5}.$$

$$= \frac{15}{82.5} = 0.181818.$$

Using Gustafson's Law,

$$S_g(200) = 0.181818 + (1 - 0.181818)(200).$$

$$= 0.181818 + 0.818182(200).$$

$$= 0.181818 + 163.6364.$$

Therefore,

$$\boxed{S_g(200) \approx 163.82}.$$

Thus, both laws agree for the same fixed-size workload.

Open/Scaled-Size Comparison

If the workload is allowed to scale while increasing the processor count from 100 to 200, the Gustafson serial fraction remains approximately

$$\beta_g = 0.1.$$

Then,

$$S_g(200) = 0.1 + (1 - 0.1)(200).$$

$$= 0.1 + 180.$$

$$= \boxed{180.1}.$$

Therefore, the results are

$$\boxed{S_g(100) = 90.1}$$

and, for 200 processors with fixed workload,

$$\boxed{S_a(200) \approx 163.82}.$$

For comparison, with an open/scaled workload,

$$\boxed{S(200) = 180.1}.$$

Task 4

Quantifying Amdahl's Formula Using Gustafson's Formula

Explain how Amdahl's formula can be quantified using Gustafson's formula. Also explain what is wrong with taking $P \to \infty$ using Gustafson's formula alone.

Relationship Between Gustafson's and Amdahl's Serial Fractions

Let

$$\beta_g$$

represent the serial fraction measured using Gustafson's approach, and let

$$\beta_a$$

represent the serial fraction used in Amdahl's Law.

For a parallel system,

$$\beta_g = \frac{T_{\text{seq}}}{T_{\text{seq}} + T_{\text{par}}}.$$

However, Amdahl's serial fraction must be measured relative to the equivalent single-processor workload.

The equivalent parallel work on one processor is

$$PT_{\text{par}}.$$

Therefore,

$$\beta_a = \frac{T_{\text{seq}}}{T_{\text{seq}} + PT_{\text{par}}}.$$

Normalize the measured parallel execution time such that

$$T_{\text{seq}} + T_{\text{par}} = 1.$$

Then,

$$T_{\text{seq}} = \beta_g$$

and

$$T_{\text{par}} = 1 - \beta_g.$$

Substituting into the Amdahl serial fraction gives

$$\beta_a = \frac{\beta_g}{\beta_g + P(1-\beta_g)}.$$

Since

$$\beta_g + P(1-\beta_g) = P - \beta_g(P-1),$$

we obtain

$$\boxed{\beta_a = \frac{\beta_g}{P - \beta_g(P-1)}}.$$

But Gustafson's speedup is

$$S_g(P) = P - \beta_g(P-1).$$

Therefore,

$$\boxed{\beta_a = \frac{\beta_g}{S_g(P)}}.$$

This relationship allows the serial percentage measured using Gustafson's approach to be converted into the serial percentage required by Amdahl's formula.

Example Using Question 1

For Question 1,

$$\beta_g = 0.2$$

and

$$S_g(10) = 8.2.$$

Therefore,

$$\beta_a = \frac{0.2}{8.2}.$$

$$\beta_a = 0.02439.$$

This is exactly the same value obtained directly from

$$\frac{10}{10 + 40(10)} = \frac{10}{410} = 0.02439.$$

What Is Wrong with Taking $P \to \infty$ in Gustafson's Formula Alone?

Gustafson's Law is

$$S_g(P) = \beta_g + (1-\beta_g)P.$$

If β_g is treated as a constant and P approaches infinity, then

$$S_g(P) \to \infty$$

as long as

$$\beta_g < 1.$$

For example, if

$$\beta_g = 0.2,$$

then

$$S_g(P) = 0.2 + 0.8P.$$

Therefore,

$$P = 10 \quad \Rightarrow \quad S_g = 8.2,$$

$$P = 30 \quad \Rightarrow \quad S_g = 24.2,$$

and as P continues to increase, the predicted speedup also continues to increase.

However, keeping β_g constant while increasing P means that the amount of parallel work is also increasing. Therefore, this corresponds to an **open or scaled-size problem**.

For a fixed-size problem, β_g cannot remain constant.

For example, in Question 1,

$$\beta_g(10) = \frac{10}{10 + 40} = 0.2.$$

With 30 processors and the same fixed workload,

$$T_{\text{par}}(30) = \frac{40}{3}.$$

Therefore,

$$\beta_g(30) = \frac{10}{10 + \frac{40}{3}} \approx 0.4286.$$

Thus,

$$\beta_g(10) = 0.2$$

but

$$\beta_g(30) \approx 0.4286.$$

As more processors are added to a fixed workload, the parallel portion becomes faster while the sequential execution time remains unchanged. Therefore, the measured sequential fraction increases.

Consequently, simply taking

$$P \to \infty$$

in Gustafson's formula while keeping β_g constant does not describe the same fixed-size problem. It implicitly assumes that the parallel workload continues to grow as processors are added.

AI Dialogue Log

This section documents the AI-assisted discussion used to understand, calculate, verify, and interpret the results of this assignment.

Dialogue 1: Understanding Amdahl's Law and Gustafson's Law

My Prompt:

> Explain Amdahl's Law and Gustafson's Law in simple terms. What is the main difference between them, especially regarding fixed problem size and scaled problem size?

AI Response:

Amdahl's Law and Gustafson's Law both estimate parallel speedup, but they view the workload differently.

Amdahl's Law is normally interpreted using a fixed-size workload. The total amount of work remains unchanged while more processors are added. The sequential portion cannot be accelerated and therefore eventually limits the speedup.

Amdahl's Law is

$$S_a(P) = \frac{1}{\beta_a + \frac{1-\beta_a}{P}},$$

where β_a is the serial fraction of the equivalent single-processor workload and P is the number of processors.

Gustafson's Law considers the serial fraction measured during parallel execution and is commonly associated with a scaled-size workload.

Gustafson's Law is

$$S_g(P) = \beta_g + (1 - \beta_g)P,$$

or equivalently,

$$S_g(P) = P - \beta_g(P - 1).$$

The main distinction is:

- Amdahl: same amount of work, more processors.
- Gustafson: more processors can be used to perform more parallel work in approximately the same execution time.

Dialogue 2: Calculating the Serial Fractions

My Prompt:

> I have a program where the sequential part takes 10 seconds and the parallel part takes 40 seconds using 10 processors. How should I calculate the serial fraction for Gustafson's Law and Amdahl's Law? Are they the same serial fraction?

AI Response:

The two serial fractions are not the same because they are normalized using different reference workloads.

For Gustafson's Law, the measured execution time is

$$10 + 40 = 50 \text{ sec.}$$

Therefore,

$$\beta_g(10) = \frac{10}{10+40} = \frac{10}{50} = 0.2.$$

For Amdahl's Law, the 40-second parallel execution on 10 processors represents

$$40 \times 10 = 400 \text{ sec}$$

of equivalent one-processor work.

Therefore, the total equivalent one-processor execution time is

$$10 + 400 = 410 \text{ sec.}$$

Thus,

$$\beta_a = \frac{10}{410} \approx 0.02439.$$

Therefore,

$$\boxed{\beta_g = 0.2}$$

while

$$\boxed{\beta_a \approx 0.02439}.$$

Dialogue 3: Checking the Baseline Speedup

My Prompt:

> Using the same measurements, calculate the speedup at the original 10 processors using both Gustafson's Law and Amdahl's Law. I want to check whether both give the same baseline result.

AI Response:

For Gustafson's Law,

$$S_g(10) = 0.2 + (1 - 0.2)(10).$$

$$= 0.2 + 0.8(10) = 8.2.$$

Therefore,

$$\boxed{S_g(10) = 8.2}.$$

For Amdahl's Law,

$$S_a(10) = \frac{1}{0.02439 + \frac{1-0.02439}{10}}.$$

$$S_a(10) \approx 8.2.$$

Therefore,

$$\boxed{S_a(10) = 8.2}.$$

Both formulas give the same baseline speedup when they are applied consistently.

Dialogue 4: Gustafson's Law for 30 Processors with Open Size

My Prompt:

> Now increase the number of processors from 10 to 30. First calculate Gustafson's Law assuming an open or scaled problem size.

AI Response:
For an open or scaled problem, the measured Gustafson serial fraction remains

$$\beta_g = 0.2.$$

For 30 processors,

$$S_g(30) = 0.2 + (1 - 0.2)(30).$$

$$= 0.2 + 0.8(30).$$

$$= 0.2 + 24.$$

Therefore,

$$\boxed{S_g(30) = 24.2}.$$

This assumes that the additional processors are used to perform a larger amount of parallel work.

Dialogue 5: Gustafson's Law for 30 Processors with Fixed Size

My Prompt:

> What happens if I use Gustafson's Law but keep the problem size fixed when going from 10 processors to 30 processors? Show how the serial fraction changes.

AI Response:
For a fixed-size problem, the same parallel work is distributed over more processors. The processor count increases by a factor of

$$\frac{30}{10} = 3.$$

Therefore,

$$T_{\text{par}}(30) = \frac{40}{3} \approx 13.333 \text{ sec.}$$

The sequential time remains

$$T_{\text{seq}} = 10 \text{ sec.}$$

The new Gustafson serial fraction is

$$\beta_g(30) = \frac{10}{10 + 13.333} \approx 0.42857.$$

Now,

$$S_g(30) = 0.42857 + (1 - 0.42857)(30).$$

$$= 0.42857 + 0.57143(30).$$

$$S_g(30) \approx 17.57.$$

Therefore,

$$\boxed{S_g(30) \approx 17.57}.$$

Dialogue 6: Amdahl's Law for 30 Processors with Fixed Size

My Prompt:

> Calculate Amdahl's Law for 30 processors assuming the workload remains fixed.

AI Response:
For the fixed-size workload, Amdahl's serial fraction remains

$$\beta_a = \frac{10}{410} \approx 0.02439.$$

Therefore,

$$S_a(30) = \frac{1}{0.02439 + \frac{1-0.02439}{30}}.$$

$$= \frac{1}{0.02439 + \frac{0.97561}{30}}.$$

$$= \frac{1}{0.05691}.$$

Therefore,

$$\boxed{S_a(30) \approx 17.57}.$$

This agrees with Gustafson's fixed-size calculation.

Dialogue 7: Amdahl's Law for an Open/Scaled Problem

My Prompt:

> Can Amdahl's Law also be used for the open or scaled-size case? Calculate it for 30 processors so I can compare it with Gustafson's result of 24.2.

AI Response:

For an open-size problem, the amount of parallel work changes. If the parallel execution still takes 40 seconds using 30 processors, then the equivalent one-processor parallel work is

$$40 \times 30 = 1200 \text{ sec.}$$

The sequential work remains

$$10 \text{ sec.}$$

Therefore,

$$\beta_a = \frac{10}{10 + 1200} = \frac{10}{1210} \approx 0.008264.$$

Now,

$$S_a(30) = \frac{1}{0.008264 + \frac{1-0.008264}{30}}.$$

$$S_a(30) \approx 24.2.$$

Therefore,

$$\boxed{S_a(30) = 24.2}.$$

Thus, Amdahl's Law and Gustafson's Law agree when the same open-size assumption is used.

Dialogue 8: Explaining the Difference Between the Predictions

My Prompt:

> Why do Amdahl's Law and Gustafson's Law sometimes appear to give different speedup predictions? Explain the difference for this example.

AI Response:

The apparent difference occurs when the two formulas are applied using different workload assumptions.

For the fixed-size workload,

$$S_a(30) \approx 17.57$$

and

$$S_g(30) \approx 17.57.$$

For the open or scaled-size workload,

$$S_a(30) = 24.2$$

and

$$S_g(30) = 24.2.$$

Therefore, the formulas are not inherently inconsistent.

The apparent difference occurs if Amdahl's Law is used for a fixed workload while Gustafson's Law is used for a scaled workload.

When the same workload assumption is used, the predictions agree.

Dialogue 9: Gustafson Speedup for 100 Processors

My Prompt:

> For Question 3, 100 processors are used, the total running time is 150 seconds, and the sequential part is 10 percent of the total running time. Calculate the Gustafson speedup step by step.

AI Response:

The sequential execution time is

$$T_{\text{seq}} = 0.10(150) = 15 \text{ sec.}$$

Therefore, the parallel execution time is

$$T_{\text{par}} = 150 - 15 = 135 \text{ sec.}$$

The Gustafson serial fraction is

$$\beta_g = \frac{15}{150} = 0.1.$$

Using Gustafson's Law,

$$S_g(100) = 0.1 + (1 - 0.1)(100).$$

$$= 0.1 + 0.9(100).$$

$$= 0.1 + 90.$$

Therefore,

$$\boxed{S_g(100) = 90.1}.$$

Dialogue 10: Converting to Amdahl's Serial Fraction

My Prompt:

> How do I convert the 15-second sequential part and 135-second parallel part on 100 processors into the serial fraction required by Amdahl's Law?

AI Response:
The parallel section takes 135 seconds using 100 processors.
The equivalent one-processor parallel work is

$$135 \times 100 = 13500 \text{ sec.}$$

The sequential portion remains

$$15 \text{ sec.}$$

Therefore, the total equivalent one-processor execution time is

$$15 + 13500 = 13515 \text{ sec.}$$

Thus, Amdahl's serial fraction is

$$\beta_a = \frac{15}{13515}.$$

$$\boxed{\beta_a \approx 0.0011099}.$$

Dialogue 11: Amdahl Prediction for 200 Processors

My Prompt:

> Using the Amdahl serial fraction from Question 3, calculate the predicted speedup when the number of processors increases from 100 to 200.

AI Response:
Using

$$\beta_a \approx 0.0011099,$$

Amdahl's Law gives

$$S_a(200) = \frac{1}{0.0011099 + \frac{1-0.0011099}{200}}.$$

$$= \frac{1}{0.0011099 + \frac{0.9988901}{200}}.$$

$$= \frac{1}{0.0011099 + 0.00499445}.$$

$$= \frac{1}{0.00610435}.$$

Therefore,

$$\boxed{S_a(200) \approx 163.82}.$$

Dialogue 12: Verifying the Result with Gustafson's Law

My Prompt:

> Can I verify the Amdahl result of about 163.82 using Gustafson's Law while keeping the workload fixed?

AI Response:
Yes.
The processor count increases from 100 to 200, so for a fixed amount of parallel work,

$$T_{\text{par}}(200) = \frac{135}{2} = 67.5 \text{ sec.}$$

The sequential execution remains

$$15 \text{ sec.}$$

The new Gustafson serial fraction is

$$\beta_g(200) = \frac{15}{15 + 67.5}.$$

$$= \frac{15}{82.5} = 0.181818.$$

Using Gustafson's Law,

$$S_g(200) = 0.181818 + (1 - 0.181818)(200).$$

$$= 0.181818 + 0.818182(200).$$

$$S_g(200) \approx 163.82.$$

Therefore,

$$\boxed{S_g(200) \approx 163.82}.$$

This verifies the Amdahl fixed-size prediction.

Dialogue 13: Relating the Two Serial Fractions

My Prompt:

> How can I calculate Amdahl's serial fraction from the serial fraction measured using Gustafson's approach? Derive the relationship.

AI Response:
Let

$$\beta_g$$

be Gustafson's serial fraction and

$$\beta_a$$

be Amdahl's serial fraction.
Gustafson's serial fraction is

$$\beta_g = \frac{T_{\text{seq}}}{T_{\text{seq}} + T_{\text{par}}}.$$

Normalize the measured parallel-machine execution time so that

$$T_{\text{seq}} + T_{\text{par}} = 1.$$

Then,

$$T_{\text{seq}} = \beta_g$$

and

$$T_{\text{par}} = 1 - \beta_g.$$

The equivalent one-processor workload is

$$T_{\text{seq}} + PT_{\text{par}}.$$

Therefore,

$$\beta_a = \frac{\beta_g}{\beta_g + P(1 - \beta_g)}.$$

Since

$$\beta_g + P(1 - \beta_g) = P - \beta_g(P - 1),$$

we obtain

$$\boxed{\beta_a = \frac{\beta_g}{P - \beta_g(P - 1)}}.$$

Since Gustafson's speedup is

$$S_g(P) = P - \beta_g(P - 1),$$

the relationship can also be written as

$$\boxed{\beta_a = \frac{\beta_g}{S_g(P)}}.$$

For Question 1,

$$\beta_g = 0.2$$

and

$$S_g(10) = 8.2.$$

Therefore,

$$\beta_a = \frac{0.2}{8.2} \approx 0.02439.$$

This agrees with the direct calculation

$$\frac{10}{410} = 0.02439.$$

Dialogue 14: Understanding the $P \to \infty$ Issue

My Prompt:

> The assignment asks what is wrong with taking P approaching infinity using Gustafson's formula alone. Explain this using the first example.

AI Response:

Gustafson's Law is

$$S_g(P) = \beta_g + (1 - \beta_g)P.$$

If β_g is kept constant and P becomes very large, the calculated speedup also becomes very large.

For example, if

$$\beta_g = 0.2,$$

then

$$S_g(P) = 0.2 + 0.8P.$$

Therefore,

$$P = 10 \Rightarrow S_g = 8.2,$$

and

$$P = 30 \Rightarrow S_g = 24.2.$$

If P continues to increase while β_g remains constant, the speedup continues to grow.

However, keeping β_g constant while increasing P means that the amount of parallel work is also increasing. Therefore, this represents an open or scaled-size problem.

For a fixed-size problem, β_g changes when P changes.

In Question 1,

$$\beta_g(10) = \frac{10}{10 + 40} = 0.2.$$

At 30 processors, the fixed-size parallel execution time becomes

$$T_{\text{par}}(30) = \frac{40}{3}.$$

Therefore,

$$\beta_g(30) = \frac{10}{10 + \frac{40}{3}} \approx 0.4286.$$

Thus,

$$\beta_g(10) = 0.2$$

but

$$\beta_g(30) \approx 0.4286.$$

Consequently, taking

$$P \to \infty$$

while keeping β_g constant should not be interpreted as infinite speedup for the same fixed-size problem. It implicitly assumes that the parallel workload continues to grow as processors are added.